

Discrete Structures

DATE: ___/___/___ Quiz 1 (Section F)

Q1: $\sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \in p$

Proof

LHS:

$$\begin{aligned} & \sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \\ \equiv & \sim(\sim p \wedge (q \vee \sim q)) \vee (p \wedge q) \\ & \text{by the distributive law.} \end{aligned}$$

$$\begin{aligned} \equiv & \sim(\sim p \wedge t) \vee (p \wedge q) \\ & \text{by the negation law for } \vee \end{aligned}$$

$$\begin{aligned} \equiv & \sim(\sim p) \vee (p \wedge q) \\ & \text{by the identity law for } \vee \end{aligned}$$

$$\begin{aligned} \equiv & p \vee (p \wedge q) \\ & \text{by the double negation law.} \end{aligned}$$

$$\equiv p \quad \text{by the absorption law.}$$

$$\equiv \text{RHS} \quad \text{Hence proved.}$$

Q2: Write statements in symbolic form

(a): Stocks are increasing but interest rates are steady
 $S \wedge i$

(b): Neither are stocks increasing nor are interest rates steady.
 $\sim S \wedge \sim i$

Q3: Use De Morgan's Law to write negations for the statement.

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(a): This computer program has a logical error in the first ten lines or it is being run with an incomplete dataset.

Let p = This computer program has a logical error in the first ten lines.

q = It is being run with an incomplete data set.

$$\boxed{p \vee q}$$

By De Morgan's Laws, The Negation is
 $\sim(p \vee q) \equiv \sim p \wedge \sim q$

(b): The dollar is at an all time high and the stock market is at record low.

Let:

p = The dollar is ^{at} all time high.

q = The stock market is at record low.

$$\boxed{p \wedge q}$$

By De Morgan's Laws, The Negation is:

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$